

Yes, I'd like to receive email about other programs from **University of Washington**.



(https://accounts.coursera.org/i/zendesk/courserahelp?return_to=https://learner.coursera.help/hc)

Yes



Bill Howe

NoSQL systems are purely about scale rather than analytics, and are arguably less relevant for the practicing data scientist. However, they occupy an important place in many practical big data platform architectures, and data scientists need to understand their

NOSQL: SYSTEMS AND CONCEPTS

Lesson 14: What problems do NoSQL systems aim to solve?



NoSQL Context and Roadmap 4 min

(</learn/data-manipulation/lecture/1r2MM/nosql-context-and-roadmap>)



NoSQL Roundup 4 min

(</learn/data-manipulation/lecture/ybTL1/nosql-roundup>)



Relaxing Consistency Guarantees 4 min

(</learn/data-manipulation/lecture/vfFvw/relaxing-consistency-guarantees>)



Two-Phase Commit and Consensus Protocols 6 min

(/learn/data-manipulation/lecture/mXS0H/two-phase-commit-and-consensus-protocols)

 Eventual Consistency 5 min

(/learn/data-manipulation/lecture/yzid6/eventual-consistency)

 CAP Theorem 4 min

(/learn/data-manipulation/lecture/lkQl0/cap-theorem)

Lesson 15: Early key-value systems and key concepts

 Types of NoSQL Systems 4 min

(/learn/data-manipulation/lecture/64vM3/types-of-nosql-systems)

 ACID, Major Impact Systems 5 min

(/learn/data-manipulation/lecture/PeQU4/acid-major-impact-systems)

 Memcached: Consistent Hashing 3 min

(/learn/data-manipulation/lecture/b3edj/memcached-consistent-hashing)

 Consistent Hashing, cont'd 5 min

(/learn/data-manipulation/lecture/fBtf7/consistent-hashing-cont-d)

 DynamoDB: Vector Clocks 5 min

(/learn/data-manipulation/lecture/TBkc1/dynamodb-vector-clocks)

 Vector Clocks, cont'd 5 min

(/learn/data-manipulation/lecture/DMI58/vector-clocks-cont-d)

Lesson 16: Document Stores and Extensible Record Stores

 CouchDB Overview 4 min

(/learn/data-manipulation/lecture/qtEKw/couchdb-overview)

 CouchB Views 4 min

(/learn/data-manipulation/lecture/8Ku98/couchb-views)

 BigTable Overview 5 min

(/learn/data-manipulation/lecture/XS5t8/bigtable-overview)

 BigTable Implementation 6 min

(/learn/data-manipulation/lecture/UEjxy/bigtable-implementation)

Lesson 17: Extended NoSQL Systems

 HBase, Megastore 4 min

(/learn/data-manipulation/lecture/Bgmsg/hbase-megastore)

 Spanner 6 min

(/learn/data-manipulation/lecture/DWeSw/spanner)

 Spanner cont'd, Google Systems 6 min

(/learn/data-manipulation/lecture/AnxaO/spanner-cont-d-google-systems)

 MapReduce-based

Help Center

Systems 5 min

(/learn/data-manipulation/lecture/ZuNyR/mapreduce-based-systems)

 Bringing Back Joins 5 min

(/learn/data-manipulation/lecture/a05wR/bringing-back-joins)

 NoSQL Rebuttal 5 min

(/learn/data-manipulation/lecture/Oy7yz/nosql-rebuttal)

Lesson 18: Pig: Programming with Relational Algebra

 Almost SQL: Pig 5 min

(/learn/data-manipulation/lecture/2bKBv/almost-sql-pig)

 Pig Architecture and Performance 3 min

(/learn/data-manipulation/lecture/uUytO/pig-architecture-and-performance)

 Data Model 4 min

(/learn/data-manipulation/lecture/pIKIS/data-model)

 Load, Filter, Group 6 min

(/learn/data-manipulation/lecture/OWn2n/load-filter-group)

 Group, Distinct, Foreach, Flatten 6 min

(/learn/data-manipulation/lecture/vjJhc/group-distinct-foreach-flatten)

Lesson 19: Pig Analytics

 CoGroup, Join 3 min

(/learn/data-manipulation/lecture/AEXyY/cogroup-join)

 Join Algorithms 4 min

(/learn/data-manipulation/lecture/HgaTK/join-algorithms)

 Skew 5 min

(/learn/data-manipulation/lecture/pzNzB/skew)

 Other Commands 3 min

(/learn/data-manipulation/lecture/SBfas/other-commands)

 Evaluation Walkthrough 4 min

(/learn/data-manipulation/lecture/m4wqm/evaluation-walkthrough)

 Review 6 min

(/learn/data-manipulation/lecture/T7an7/review)

Lesson 20: Spark

 Context 3 min

(/learn/data-manipulation/lecture/WsGZ0/context)

 Spark Examples 5 min

(/learn/data-manipulation/lecture/2Quex/spark-examples)

 RDDs, Benefits 6 min

(/learn/data-manipulation/lecture/Tsxta/rdds-benefits)



Bill Howe

Graph-structured data are increasingly common in data science contexts due to their ubiquity in modeling the communication between entities: people (social networks), computers (Internet communication), cities and countries (transportation networks), or

GRAPH ANALYTICS

Lesson 21: Structural Tasks

 Graph Overview 6 min

(/learn/data-manipulation/lecture/90Fbl/graph-overview)

 Structural Analysis 4 min

(/learn/data-manipulation/lecture/wYFHI/structural-analysis)

 Degree Histograms,
Structure of the
Web 4 min

(/learn/data-manipulation/lecture/H4dnh/degree-histograms-structure-of-the-web)

 Connectivity and
Centrality 5 min

(/learn/data-manipulation/lecture/ebVxC/connectivity-and-centrality)

Lesson 22: Traversal Tasks

 PageRank 4 min

(/learn/data-manipulation/lecture/iK7MT/pagerank)

 PageRank in more
Detail 3 min

(/learn/data-manipulation/lecture/yUYMt/pagerank-in-more-detail)

 Traversal Tasks: Spanning



Trees and Circuits 5 min

(/learn/data-manipulation/lecture/8DHEC/traversal-tasks-spanning-trees-and-circuits)



Traversal Tasks:
Maximum Flow 2 min

(/learn/data-manipulation/lecture/tzklW/traversal-tasks-maximum-flow)

Lesson 23: Pattern Matching Tasks and Graph Query



Pattern Matching 6 min

(/learn/data-manipulation/lecture/SKX5D/pattern-matching)



Querying Edge
Tables 4 min

(/learn/data-manipulation/lecture/BYS2b/querying-edge-tables)



Relational Algebra and
Datalog for Graphs 5 min

(/learn/data-manipulation/lecture/U8zVV/relational-algebra-and-datalog-for-graphs)



Querying Hybrid
Graph/Relational
Data 4 min

(/learn/data-manipulation/lecture/BZg5u/querying-hybrid-graph-relational-data)



Graph Query Example:
NSA 6 min

(/learn/data-manipulation/lecture/R57UJ/graph-query-example-nsa)

Lesson 24: Recursive

Queries

-  Graph Query Example:
Recursion 4 min

(/learn/data-manipulation/lecture/OMStu/graph-query-example-recursion)

-  Evaluation of Recursive
Programs 4 min

(/learn/data-manipulation/lecture/u0Jus/evaluation-of-recursive-programs)

-  Recursive Queries in
MapReduce 5 min

(/learn/data-manipulation/lecture/U1F9Q/recursive-queries-in-mapreduce)

-  The End-Game
Problem 3 min

(/learn/data-manipulation/lecture/D2VRq/the-end-game-problem)

Lesson 24: Representations and Algorithms

-  Representation: Edge
Table, Adjacency
List 4 min

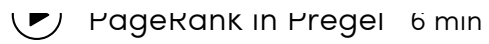
(/learn/data-manipulation/lecture/a17wQ/representation-edge-table-adjacency-list)

-  Representation:
Adjacency Matrix 3 min

(/learn/data-manipulation/lecture/iNz0u/representation-adjacency-matrix)

-  PageRank in
MapReduce 5 min

(/learn/data-manipulation/lecture/IOYBs/pagerank-in-mapreduce)



(/learn/data-manipulation/lecture/uxVhd/pagerank-in-pregel)